

International Mathematical Olympiad Problems (1986–1988)

1986/1

Let d be any positive integer not equal to 2, 5, or 13. Show that one can find distinct a, b in the set $\{2, 5, 13, d\}$ such that $ab - 1$ is not a perfect square.

1986/2

A triangle $A_1A_2A_3$ and a point P_0 are given in the plane. Define $A_s = A_{s-3}$ for all $s \geq 4$. Construct points $P_1, P_2, \dots$ such that P_{k+1} is the image of P_k under a rotation with center A_{k+1} through angle 120° clockwise. Prove that if $P_{1986} = P_0$, then the triangle $A_1A_2A_3$ is equilateral.

1986/3

To each vertex of a regular pentagon an integer is assigned so that the sum of all five numbers is positive. If three consecutive vertices are assigned x, y, z with $y < 0$, then the operation $(x, y, z) \mapsto (x + y, -y, z + y)$ is allowed. Determine whether this procedure necessarily ends after finitely many steps.

1986/4

Let A, B be adjacent vertices of a regular n -gon ($n \geq 5$) centered at O . A triangle XYZ , congruent to OAB , moves so that Y, Z trace the boundary of the polygon while X remains inside. Find the locus of X .

1986/5

Find all functions $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that:

$$f(xf(y))f(y) = f(x + y), \quad f(2) = 0, \quad f(x) > 0 \text{ for } 0 \leq x < 2.$$

1986/6

Given a finite set of integer-coordinate points in the plane, is it always possible to color them red/white so that for any line parallel to an axis, the difference between counts of red and white points is at most 1?

1987/1

Let $P_n(k)$ be the number of permutations of $\{1, \dots, n\}$ with exactly k fixed points. Prove:

$$\sum_{k=0}^n k \cdot P_n(k) = n!.$$

1987/2

In an acute triangle ABC , the bisector of $\angle A$ meets BC at L and the circumcircle again at N . From L , draw perpendiculars to AB, AC with feet K, M . Prove that quadrilateral $AKNM$ and triangle ABC have equal areas.

1987/3

Let $x_1, \dots, x_n$ be reals with $x_1^2 + \dots + x_n^2 = 1$. Prove that for every integer $k \geq 2$, there exist integers $a_1, \dots, a_n$, not all zero, with $|a_i| < k - 1$ and

$$|a_1x_1 + \dots + a_nx_n| < \frac{(k-1)\sqrt{n}}{kn-1}.$$

1987/4

Prove that no function $f: \mathbb{Z}_{\geq 0} \rightarrow \mathbb{Z}_{\geq 0}$ satisfies $f(f(n)) = n + 1987$.

1987/5

Prove that for any integer $n \geq 3$, there exists a set of n points in the plane such that distances are irrational, yet every triple forms a triangle of rational area.

1987/6

Let $n \geq 2$. Prove that if $k^2 + k + n$ is prime for all integers $0 < k \leq \sqrt{n}/3$, then it is prime for all $0 \leq k \leq n - 2$.

1988/1

Two concentric circles of radii $R > r$. Point P on smaller circle, point B on larger circle. Line BP meets larger circle again at C . Perpendicular at P meets smaller circle again at A . (i) Find values of $BC^2 + CA^2 + AB^2$. (ii) Find locus of midpoint of BC .

1988/2

Let $A_1, \dots, A_{2n+1}$ be subsets of a set B with: (a) Each has $2n$ elements. (b) Each pair intersects in exactly one element. (c) Every element of B belongs to at least two subsets. For which n can one assign 0/1 to elements so that each A_i has exactly n zeros?

1988/3

Define f on positive integers by: $f(1) = 1, f(3) = 3, f(2n) = f(n), f(4n+1) = 2f(2n+1) - f(n), f(4n+3) = 3f(2n+1) - 2f(n)$. Determine number of $n \leq 1988$ with $f(n) = n$.

1988/4

Show that the set of real x satisfying

$$\sum_{k=1}^{70} \frac{x-k}{k^5} \geq 4$$

is a union of disjoint intervals whose total length is 1988.

1988/5

In right triangle ABC ($\angle A = 90^\circ$), let D be foot of altitude from A . The line joining incenters of ABD, ACD meets AB, AC at K, L . If S, T are areas of ABC, AKL , show $S \geq 2T$.

1988/6

Let a, b be positive integers with $ab+1 \mid a^2+b^2$. Show that

$$\frac{a^2+b^2}{ab+1}$$

is a perfect square.